DESENVOLVIMENTO INTEGRADO DE SOFTWARE

2023/24

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Universidade do Minho
Escola de Engenharia

DADOS GERAIS

- **Aulas**
  - **T**: 3.ª-f ; 09h10-11h00, presencial (Francisco Duarte)
  - PL: 2.ª-f: 15h10-19h00, presencial (Paula Monteiro)
- **Avaliação**:
  - M_Inception: 10%
  - M_Elaboration: 15%
  - M_Construction: 30%
  - M_Transition: 25%
  - 4x TBL: 20%

Todos os momentos de avaliação: **nota mínima >= 8/20**

- **Grupos**: vários grupos por aula PL, papéis pré-estabelecidos

Referênciais:

- Software Engineering Body of Knowledge **SWEBoK** www.swebok.org
- Processo de Desenvolvimento de Software **OpenUp** https://download.eclipse.org/technology/epf/OpenUP/published/openup_published_1.5.1.5_20121212/openup/index.htm

- **Plataforma de ensino**: eLearning UMinho https://elearning.uminho.pt

DADOS GERAIS: PLANEAMENTO

semana	2ª Feira	DIS-PL (3 grupos) segunda-feira, 15h-17h, Ed11 - 1.39	3ª Feira	DIS-T terça-feira, 09h-11h, Ed11 - 0.08
05/02 a 10/02	semana apresentação T		DIS/T01	Apresentação, Plano, Avaliações: cliente e parceiro, roles OpenUP, OpenUP Inception Disciplina Environment; Project Libre, Risk Analysis; SWEBOK
12/02 a 17/02	DIS/PL01	definição roles, explicitação trabalho, plano inicial; definição inicial de ferramentas e tecnologias suporte (drive cloud, Visual Paradigm, repo dados, etc…) Planeamento do Projeto, Análise de Riscos	Carnaval	
19/02 a 24/02	DIS/PL02	Visita (virtual?) cliente: TUB (EUB- Estacionamentos Urbanos de Braga) Exercício: planeamento com Project Libre e OpenUP; Trabalho projecto: Planeamento com Project Libre e OpenUP	DIS/T02	UML Use Cases; UML: Sequências; Requisitos: FURPS+
26/02 a 02/03	DIS/PL03	M_Inception Apresentação Aula, Entrega: +24h Exercício: UML Use Case e Sequências (Pizzaria); Trabalho projecto: requisitos funcionais e não funcionais com UC, Sequências, e FURPS+	DIS/T03	UML: Classes; OpenUP: Elaboration
04/03 a 09/03	DIS/PL04	**Team-Based Learning 1: Classes** Trabalho projecto: Classes	DIS/T04	UML: Actividades; UML: Estados
11/03 a 16/03	DIS/PL05	**Team-based Learning 2: Actividades** exercício: UML Estados, UML: Deployment (Pizzaria); trabalho de projectos: estados, actividades, deployment	DIS/T05	Visita (virtual?) parceiro: Seminário: Neadvance NIOP; UML: Deployment
18/03 a 23/03	DIS/PL06	M_Elaboration Apresentação Aula, Entrega: +24h	DIS/T06	Apresentação protótipo 1 ao docente OpenUP: Constuction;
25/03 a 30/03	Férias Páscoa		Férias Páscoa	
01/04 a 06/04	Férias Páscoa		DIS/T07	Arquitetura de Software Architectural Patterns;
08/04 a 13/04	DIS/PL07	trabalho de projeto: Construction	DIS/T08	**TBL3: Modelos de Processos de Desenvolvimento de Software**
15/04 a 20/04	DIS/PL08	trabalho de projeto: Construction	DIS/T09	Conceção de Software Design Patterns
22/04 a 27/04	DIS/PL09	M_Construction Apresentação Aula, Entrega: +24h	DIS/T10	Apresentação produto final ao docente OpenUP: Transition;
29/04 a 04/05	DIS/PL10	trabalho de projeto: Transition	DIS/T11	**TBL4: Teste de Software**
06/05 a 11/05	Enterro da gata		Enterro da gata	
13/05 a 18/05	DIS/PL11	trabalho de projeto: Transition	DIS/T12	Preparação apresentação: Pitch com Business Model Canvas TBL
20/05 a 25/05	DIS/PL12	trabalho de projecto: Transition; Preparação apresentação M_Transition	DIS/T13	M_Transition Apresentação ao cliente e parceiro e docentes, Entrega: +24h

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OBJETIVOS DE APRENDIZAGEM

1. Perceber a **engenharia de software**: usar **SWEBoK** v3.0 – V4.0 draft (aulas T);

1. Perceber os diferentes tipos de **processos de desenvolvimento de software** e perceber o **OpenUp**;

1. Ser capaz de **criar soluções completas** de software, baseadas em **modelos**;

1. Ter contacto com distintos **paradigmas de concepção** (design) e de **arquiteturas de software**;

1. Saber usar **ferramentas auxiliares no desenvolvimento de software** (e.g. planeamento, repositórios de dados, IDE, ...) ;

1. Ser introduzidos aos paradigmas **No Code/Low Code** e usá-los em projecto.

0: Ética

Regra 1

Todos os membros de um grupo são responsáveis por todo o conteúdo dos trabalhos de grupo.

Regra 2

Em qualquer avaliação, cada aluno deve honestamente informar qualquer ajuda recebida e fontes de informação usadas.

Regra 3

Nas avaliações orais, cada aluno deve ser capaz de apresentar e responder a questões sobre qualquer parte do trabalho de grupo

PROJECTO

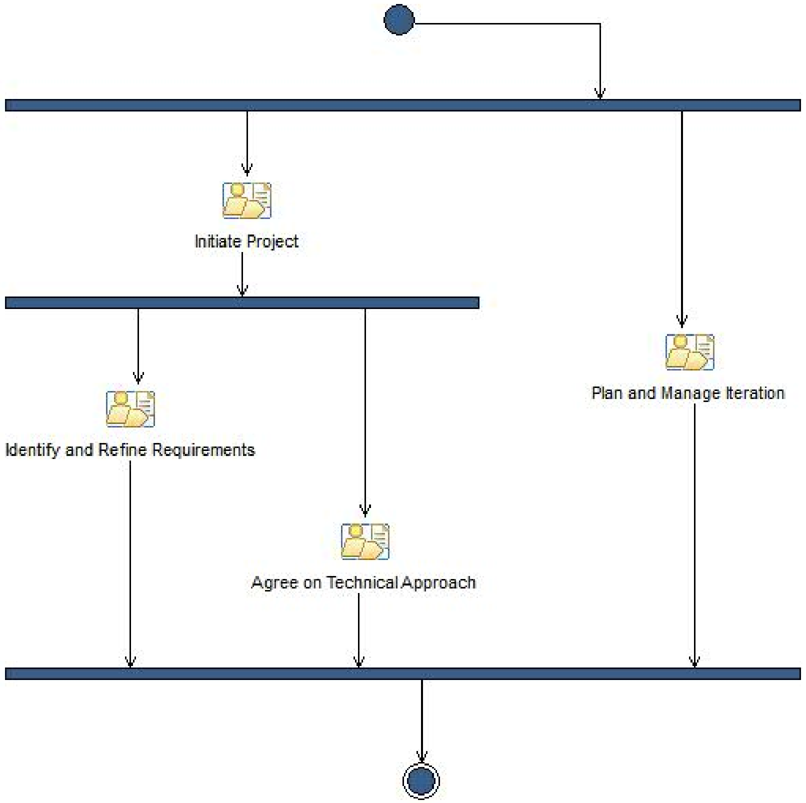
Os momentos de avaliação:

M_Inception, M_Elaboration, M_Construction, M_Transition

São baseado em trabalho de projecto, seguindo o processo OpenUp, para o **cliente**

OpenUP: Inception

2. OpenUP: Inception



Iteração na fase *Inception*

Initiate Project:
-Develop Technical Vision
-Plan Project

Identify and Refine Requirements
-Identify and Outline Requirements
-Detail Use-Case Scenarios
-Detail System-Wide Requirements
-Create Test Cases

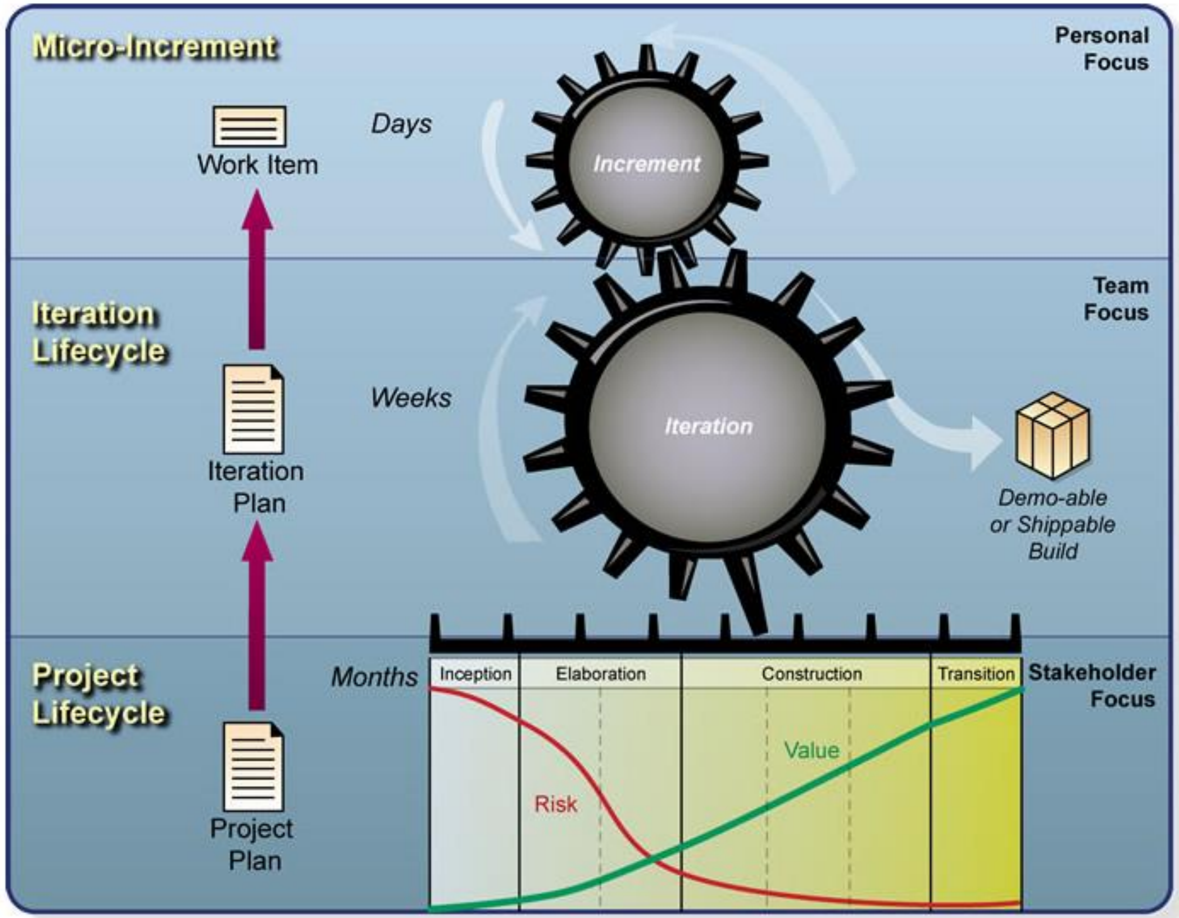
Agree on Technical Approach
Envision the Architecture

Plan and Manage Iteration
Plan Iteration
Prepare Environment
Manage Iteration
Assess Results

Tailor the Process
Set Up Tools
Verify Tool Configuration and Installation
Deploy the Process

OpenUP online

OpenUP: Roles



Roles para o trabalho de grupo:

Role Set: Basic Roles

This is the list of basic software development roles.

Relationships

Roles	• Analyst • Any Role • Architect • Developer • Project Manager • Stakeholder • Tester

Role Set: Environment

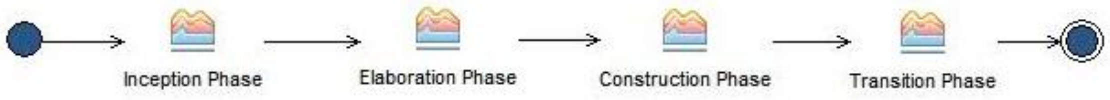
This standard category lists all environment roles in software development.

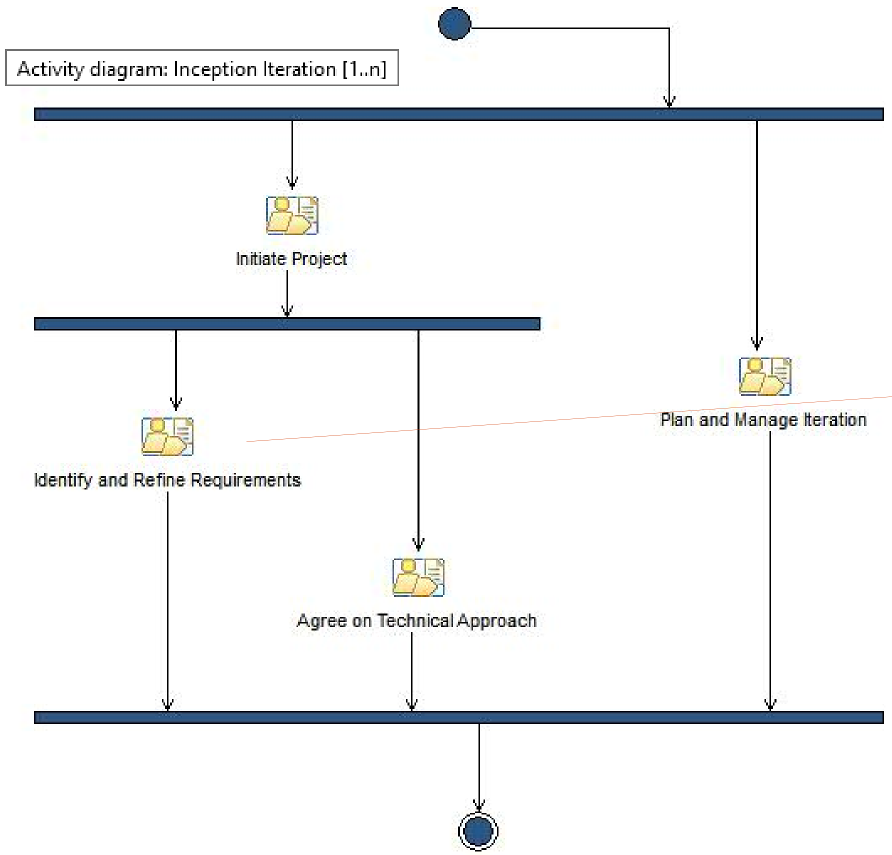
Relationships

Roles	• Process Engineer • Tool Specialist

PROCESSO OPENUP

Roles para o trabalho de grupo:





Breakdown Element
Identify and Outline Requirements
Detail Use-Case Scenarios
Detail System-Wide Requirements
Create Test Cases

Task: Identify and Outline Requirements

This task describes how to identify and outline the requirements for the system so that the scope of work can be determined.

Purpose

The purpose of this task is to identify and capture functional and non-functional requirements for the system. These requirements form the basis of communication and agreement between
understand the requirements at a high-level so that the initial scope of work can be determined. Further analysis will be performed to detail these requirements prior to implementation.

Relationships

Roles	Primary: • Analyst	Additional: • Architect • Developer • Stakeholder • Tester
Inputs	Mandatory: • None	Optional: • [Technical Specification]
Outputs	• Glossary • System-Wide Requirements • Use Case • Use-Case Model • Work Items List	

Steps

Properties

More Information

Concepts	• Requirements • System-Wide Requirements
Guidelines	• Developing System-Wide Requirements Specification • Effective Requirement Reviews • Identify and Outline Actors and Use Cases • Requirements Gathering Techniques

RUP – DISCIPLINA DE ENVIRONMENT

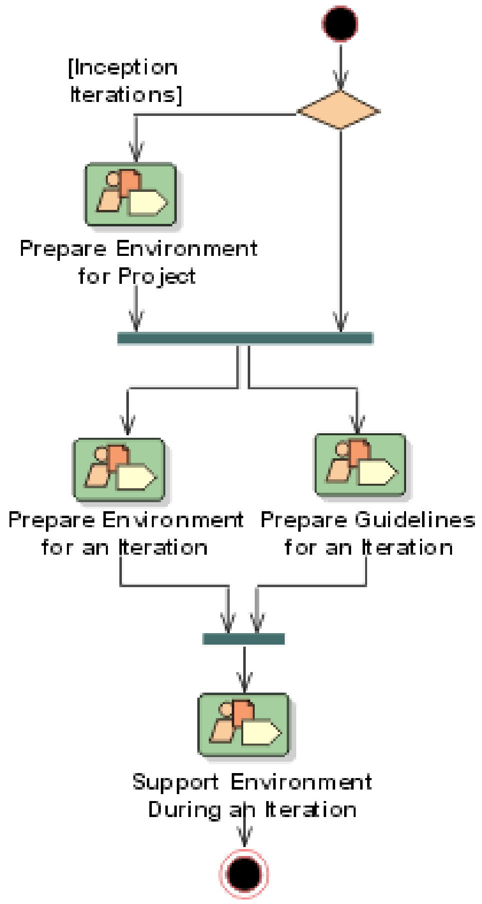
AMBIENTE DE DESENVOLVIMENTO

RUP: ENVIRONMENT

O **ambiente de desenvolvimento** para um projecto de desenvolvimento de software é o conjunto de todas as coisas que o projecto necessita para desenvolver e entregar o sistema, como

- Ferramentas
- Guidelines
- Processo
- Templates
- Infraestrutura

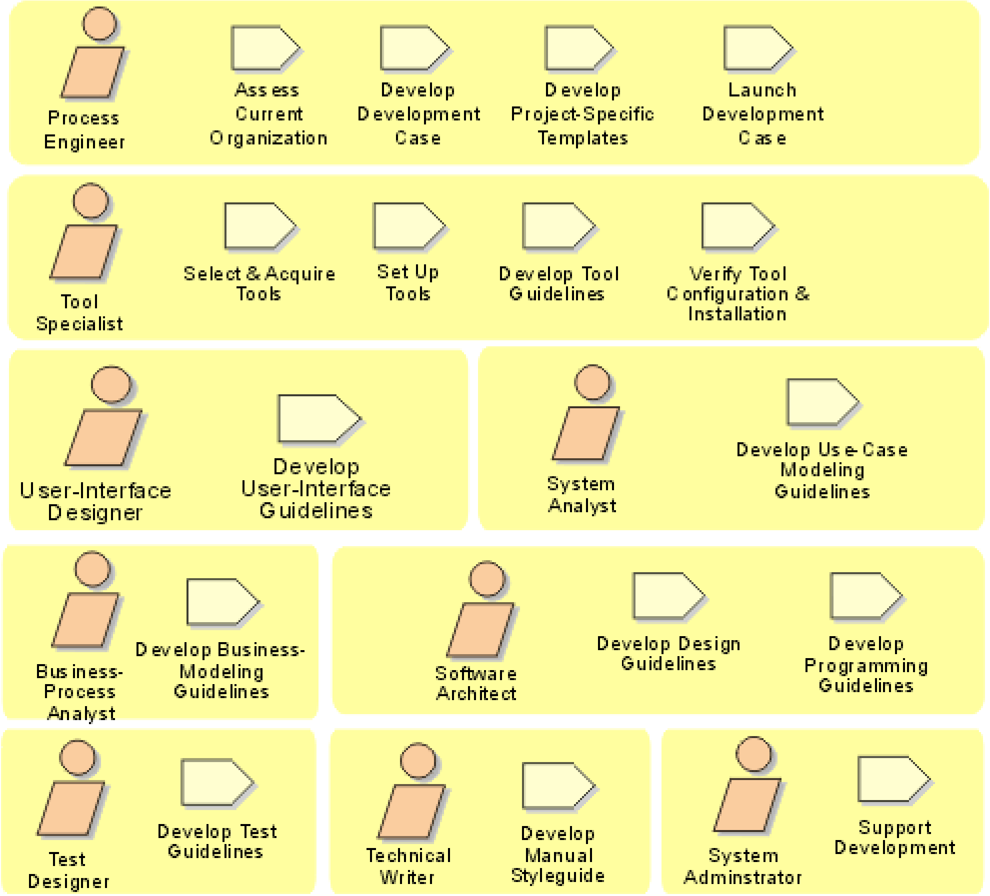
Fluxo de trabalho



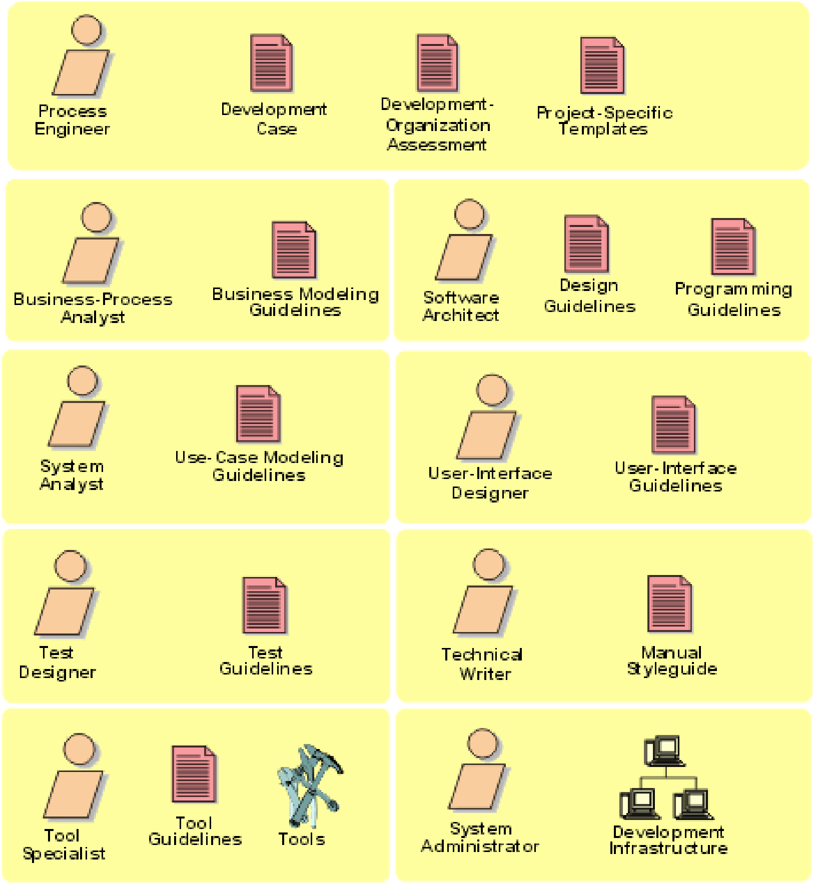
AMBIENTE

RUP: ENVIRONMENT

Vista geral sobre as actividades



Vista geral sobre os artefactos



HTTPS://SCEWEB.UHCL.EDU/HELM/RATIONALUNIFIEDPROCESS/

SWEBOK

ENGENHARIA DE SOFTWARE

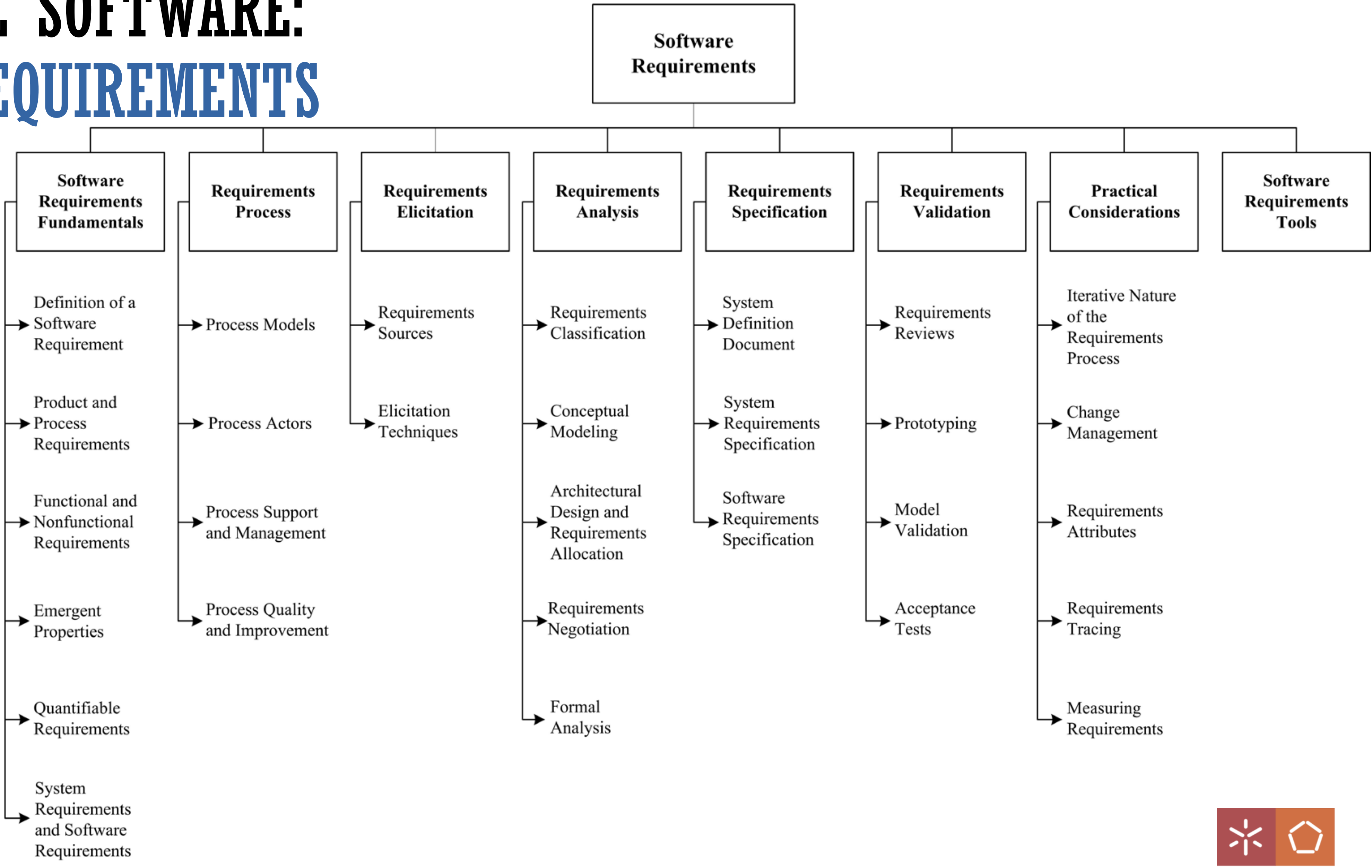
ISO/IEC/IEEE Systems and Software Engineering Vocabulary (SEVOCAB):

"the application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software"

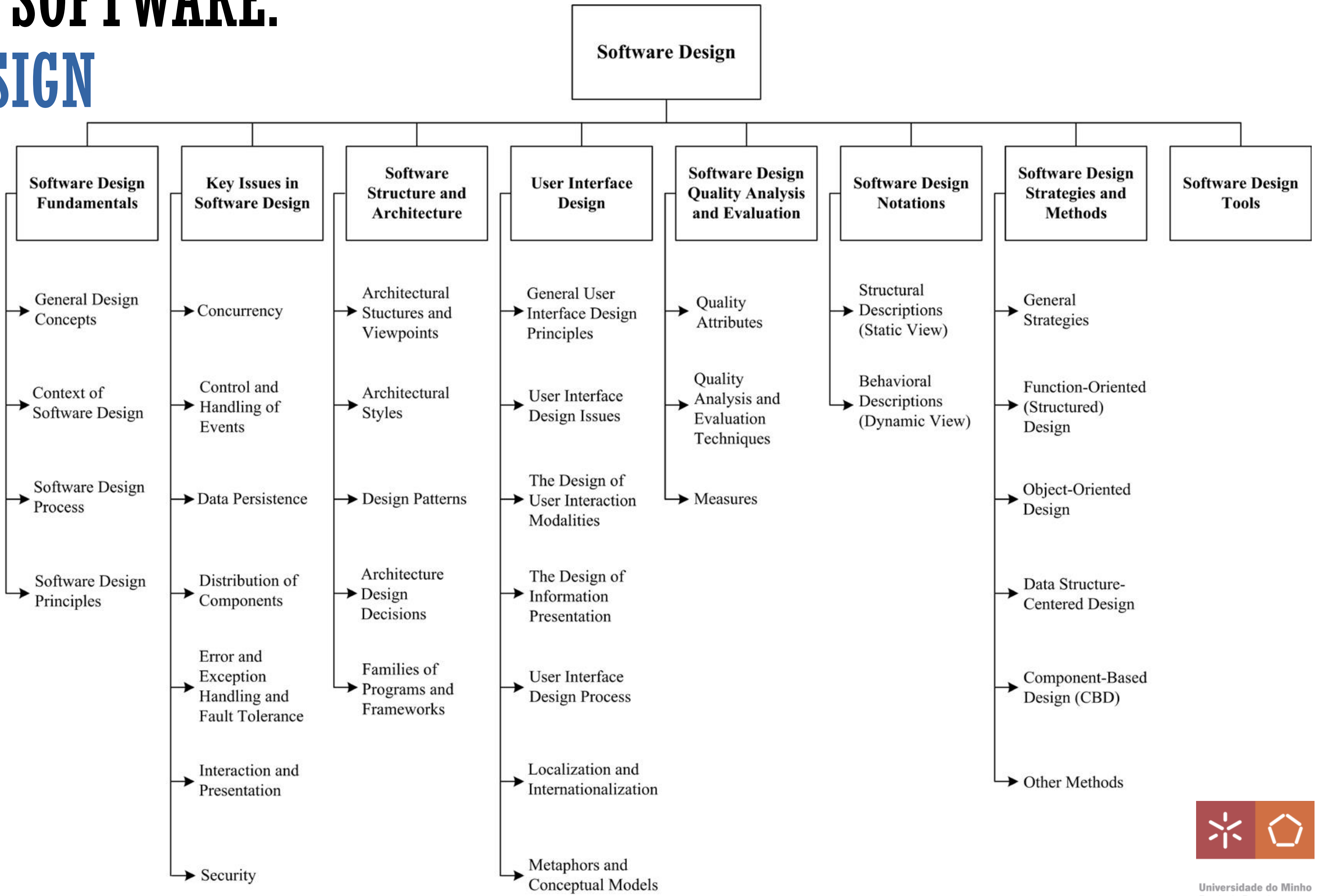
ENGENHARIA DE SOFTWARE: SWEBOK KNOWLEDGE AREAS

Table I.1. The 15 SWEBOK KAs
Software Requirements
Software Design
Software Construction
Software Testing
Software Maintenance
Software Configuration Management
Software Engineering Management
Software Engineering Process
Software Engineering Models and Methods
Software Quality
Software Engineering Professional Practice
Software Engineering Economics
Computing Foundations
Mathematical Foundations
Engineering Foundations

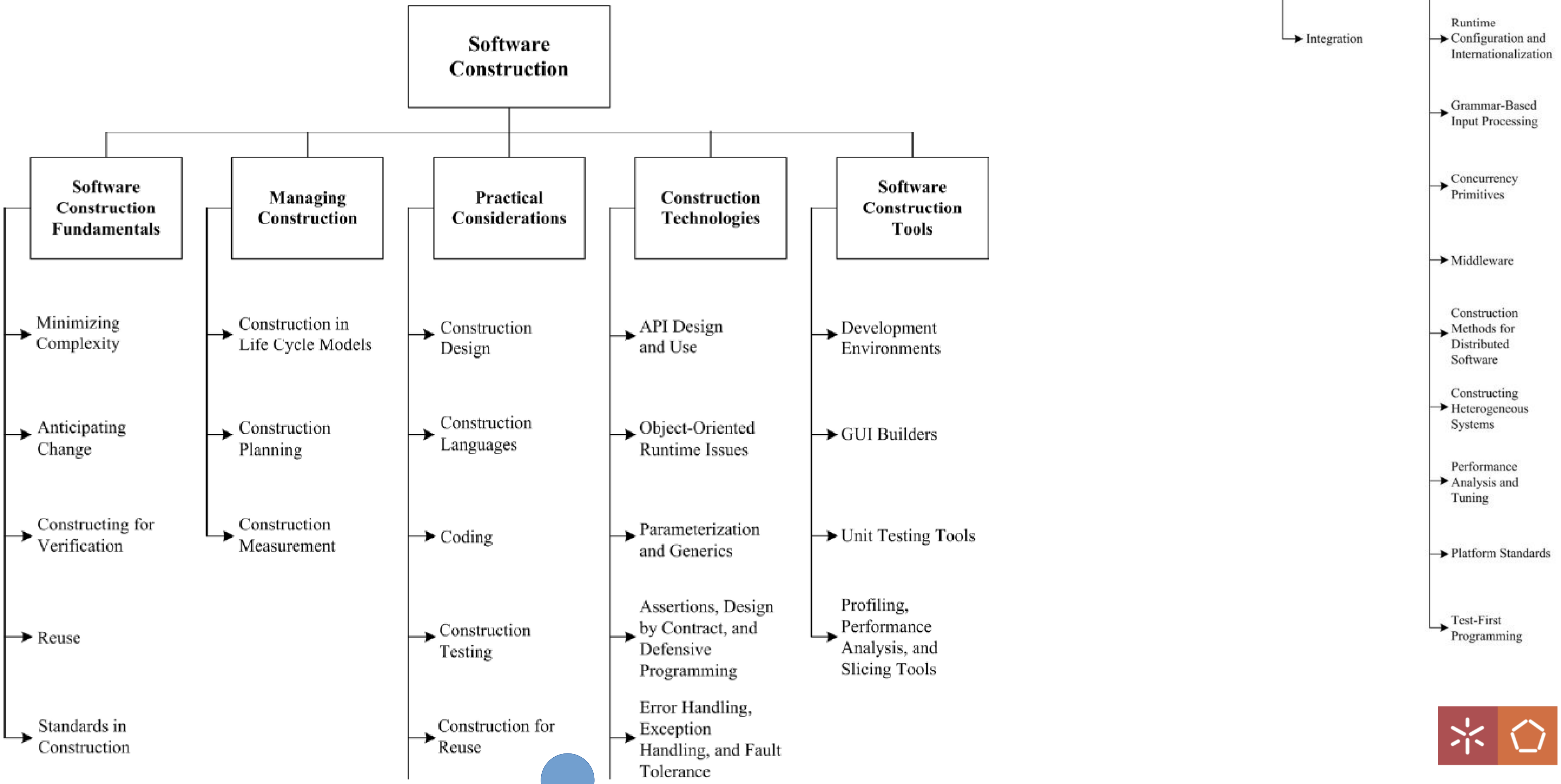
ENGENHARIA DE SOFTWARE:
1. SOFTWARE REQUIREMENTS



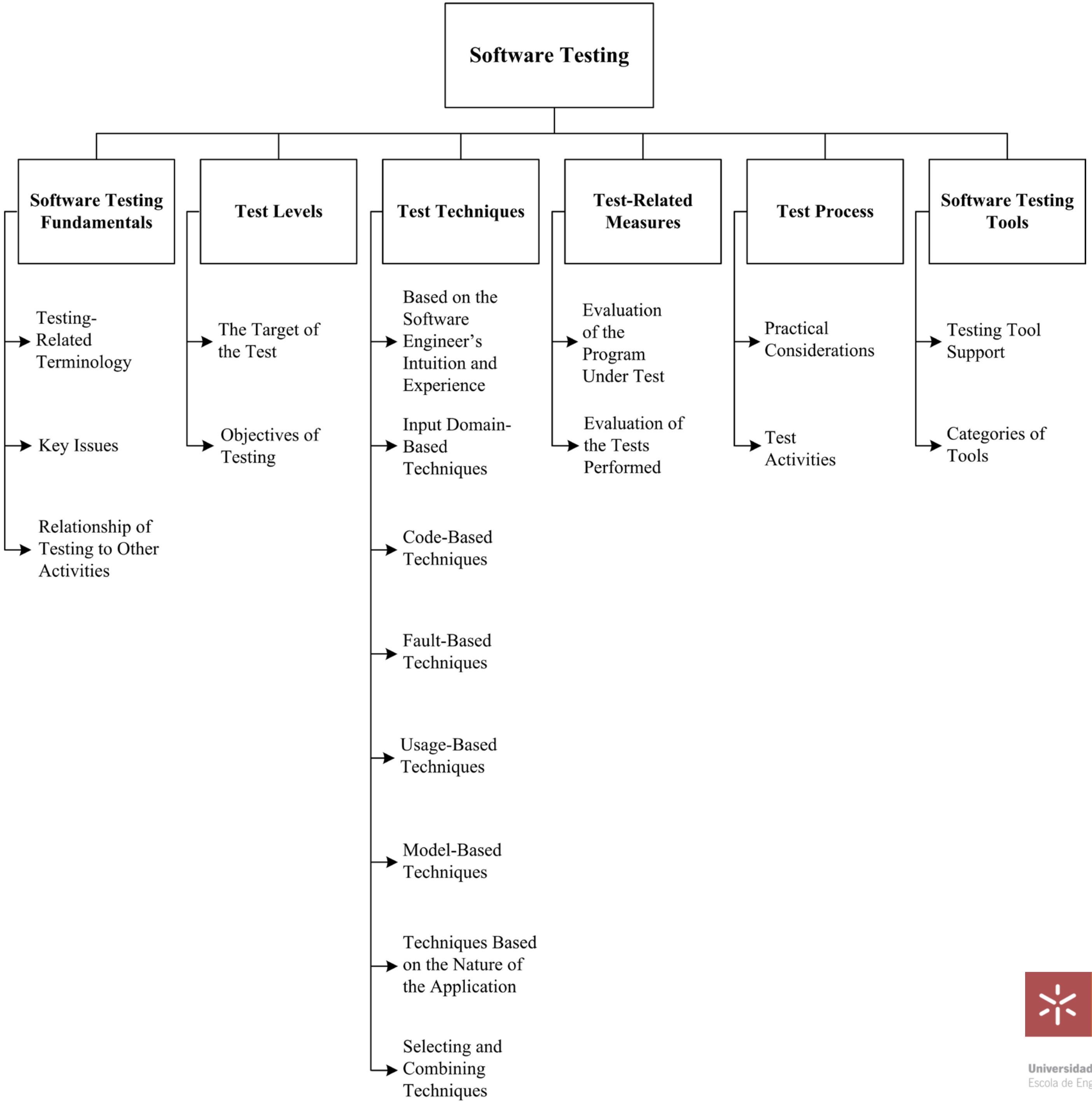
ENGENHARIA DE SOFTWARE:
2. SOFTWARE DESIGN



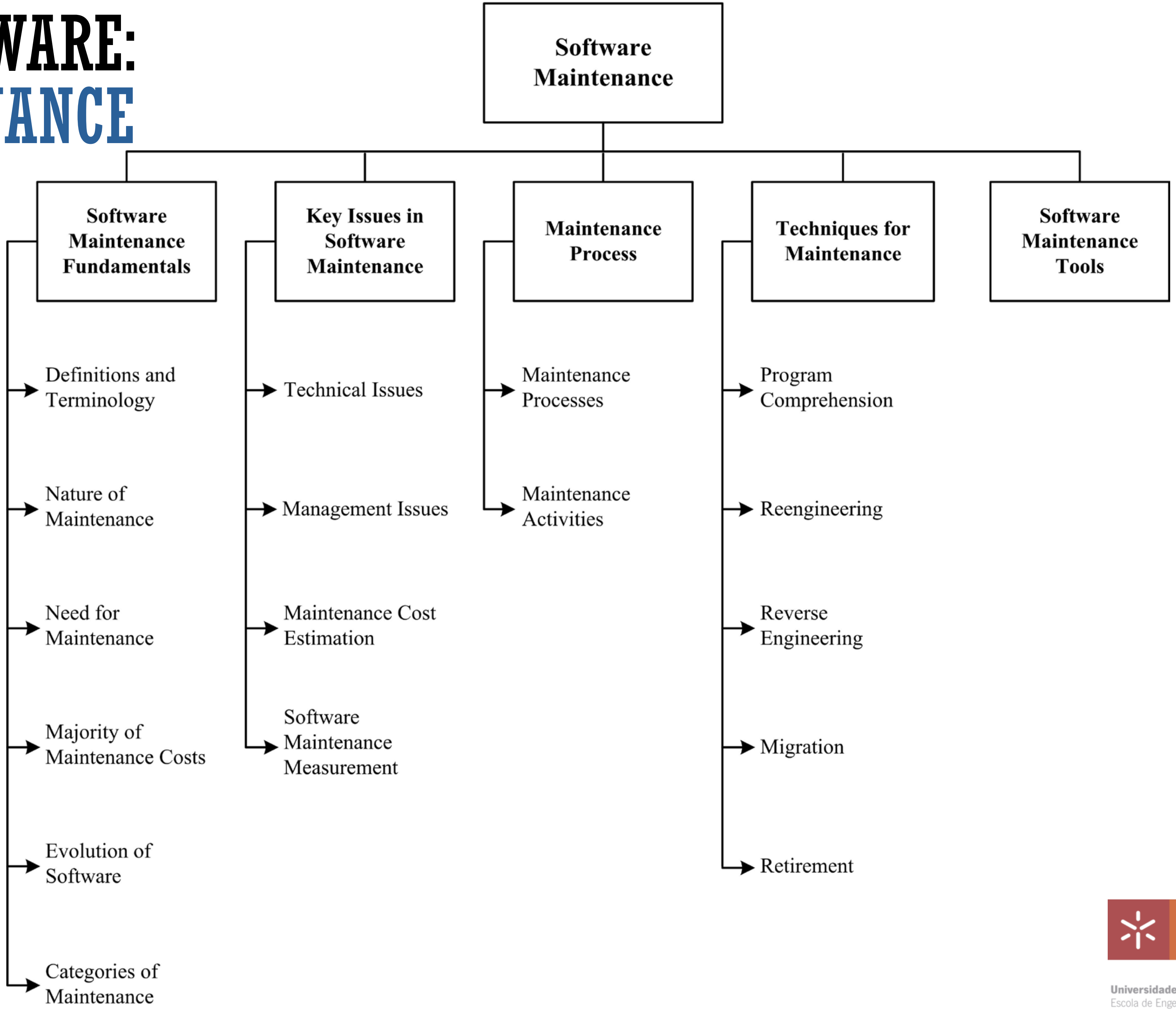
ENGENHARIA DE SOFTWARE:
3. SOFTWARE CONSTRUCTION



ENGENHARIA DE SOFTWARE:
4. SOFTWARE TESTING



ENGENHARIA DE SOFTWARE:
5. SOFTWARE MAINTENANCE



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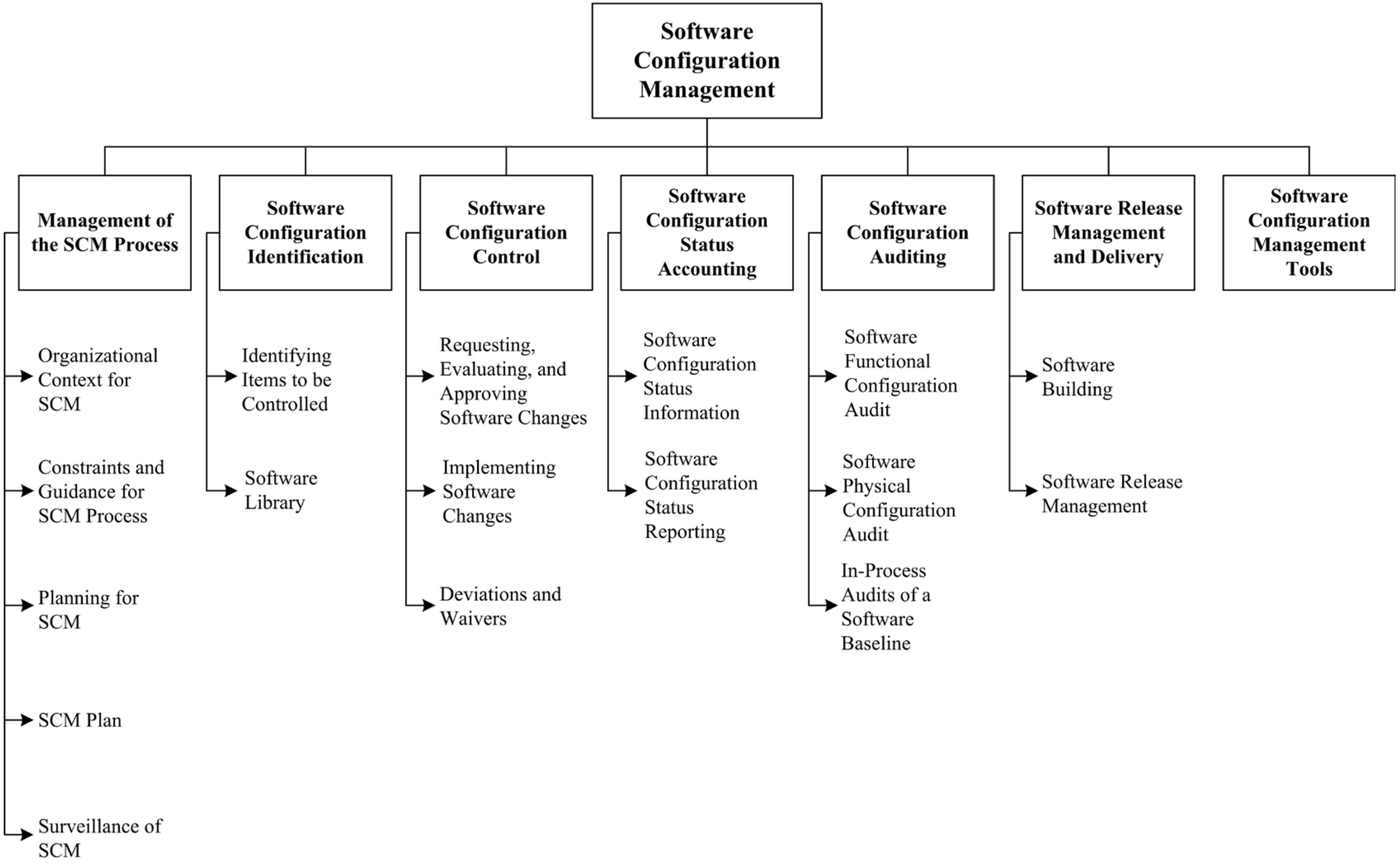
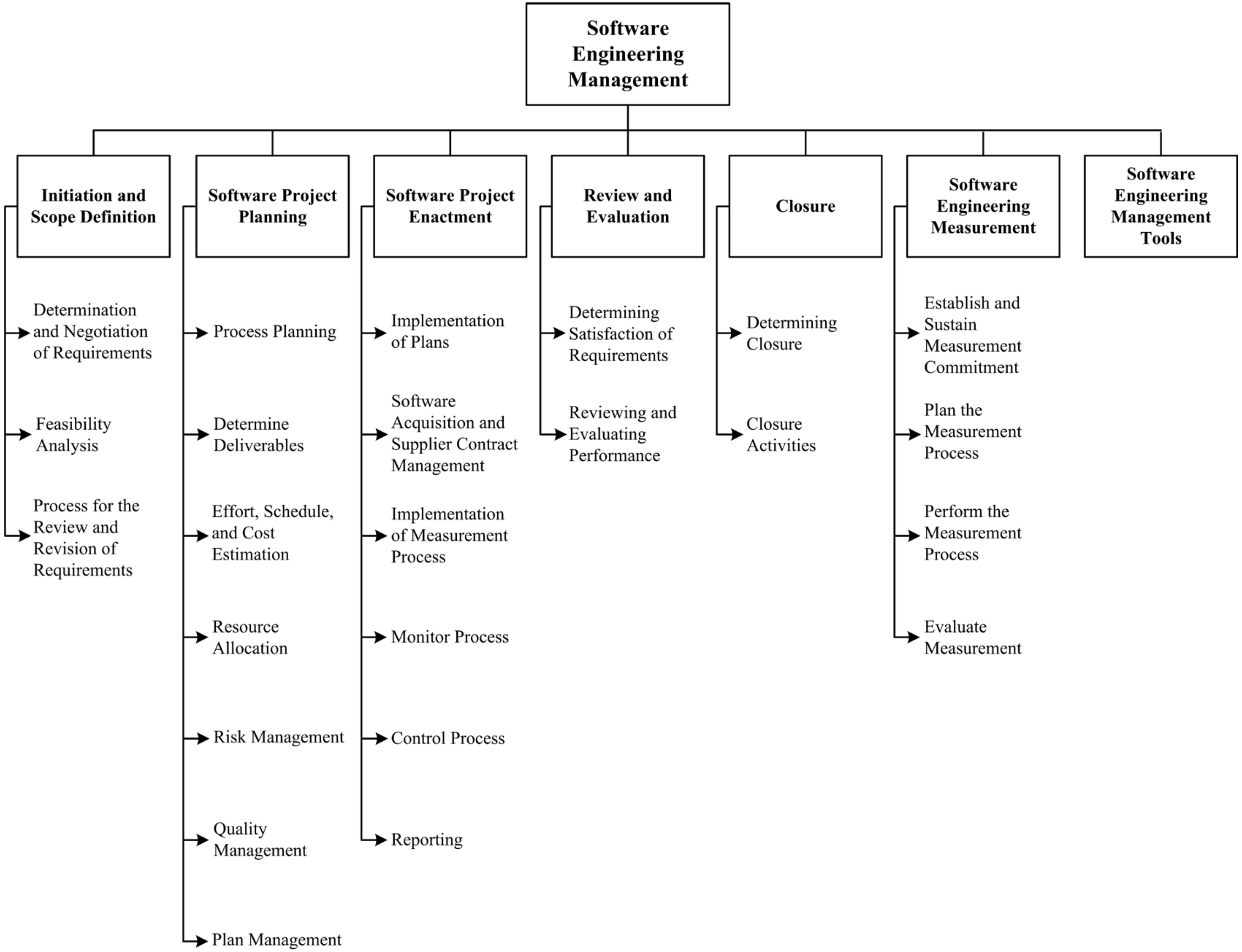
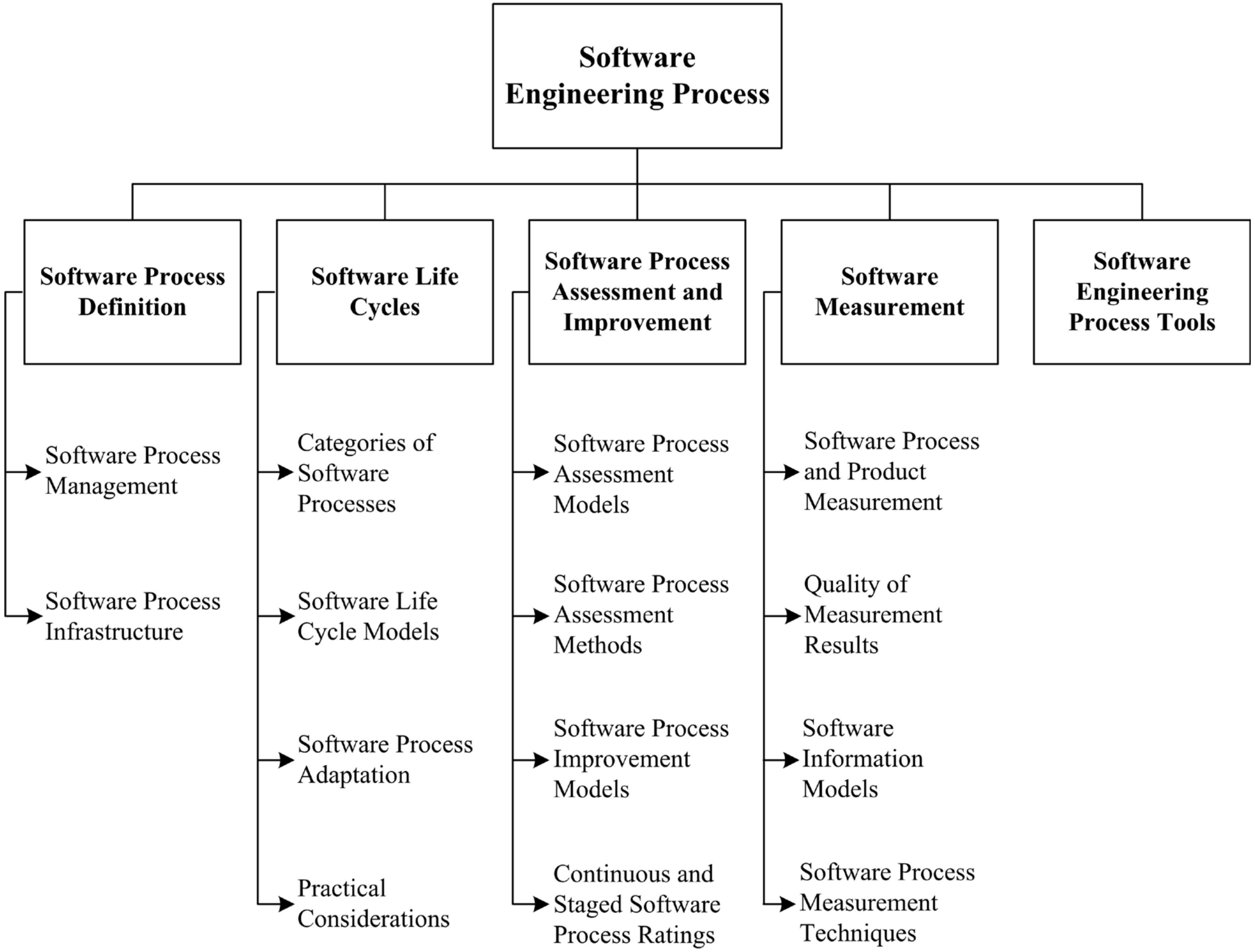
ENGENHARIA DE SOFTWARE:
6. SOFTWARE CONFIGURATION MANAGEMENT



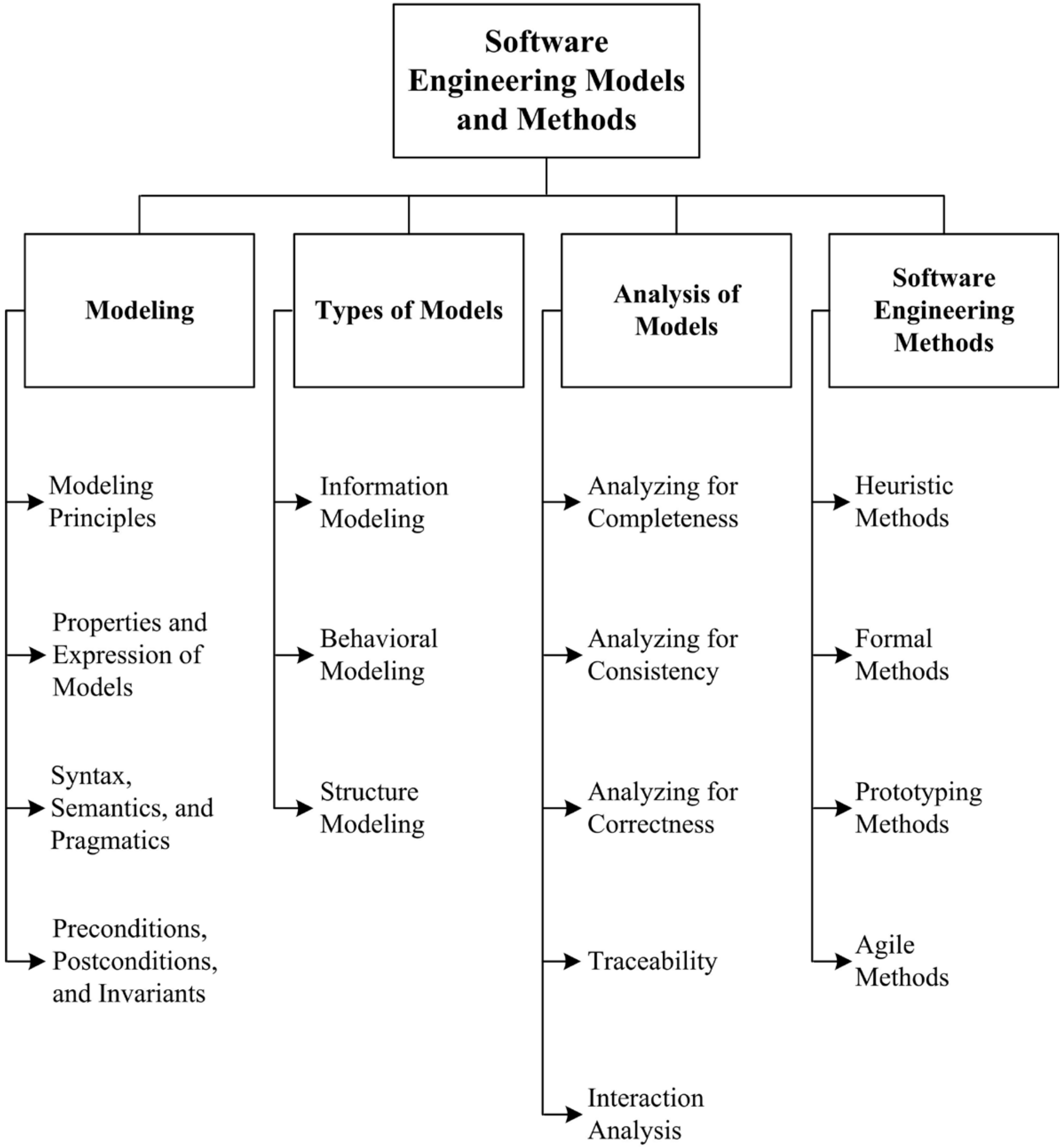
ENGENHARIA DE SOFTWARE:
7. SOFTWARE ENGINEERING MANAGEMENT



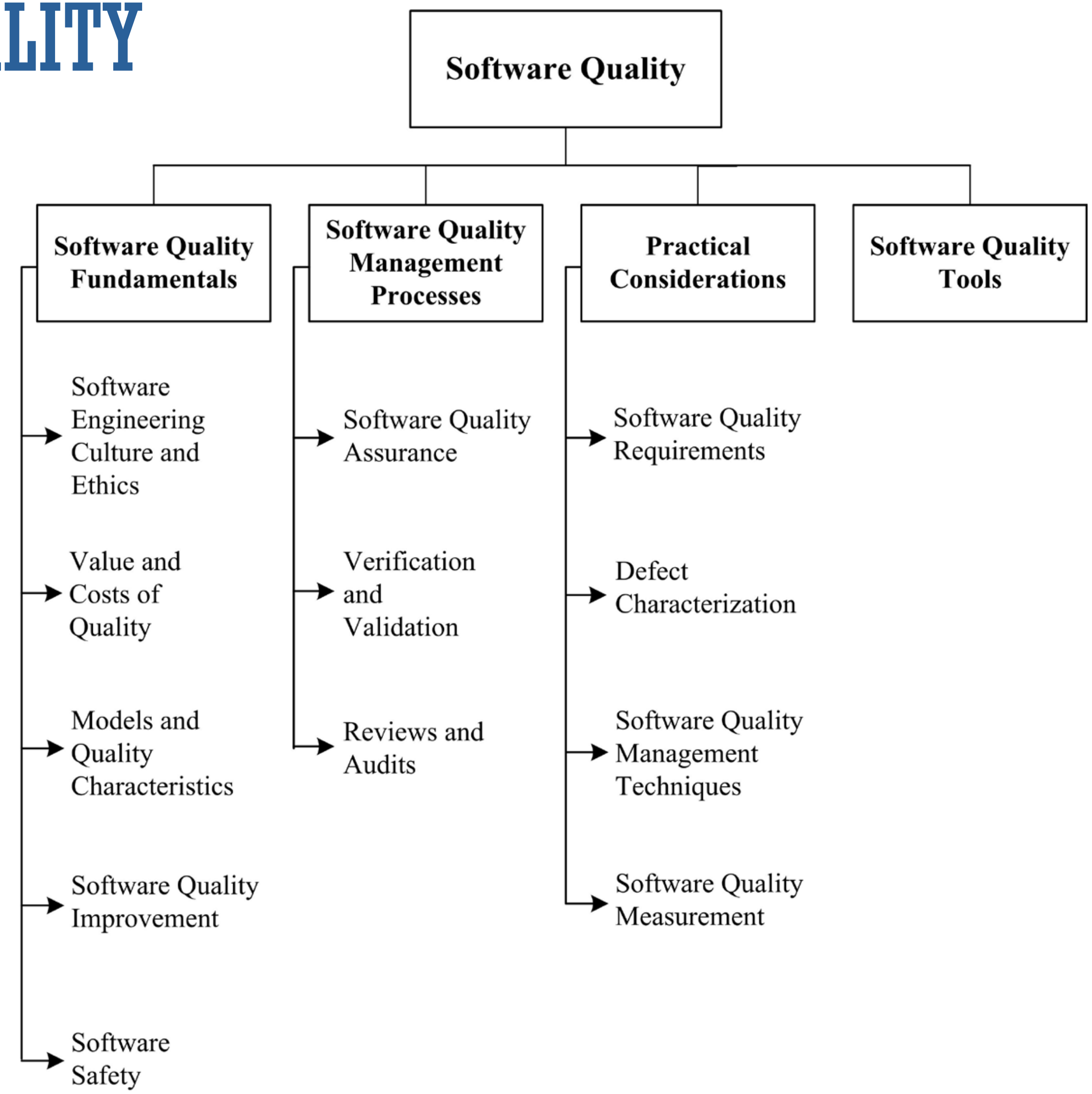
ENGENHARIA DE SOFTWARE:
8. SOFTWARE ENGINEERING PROCESS



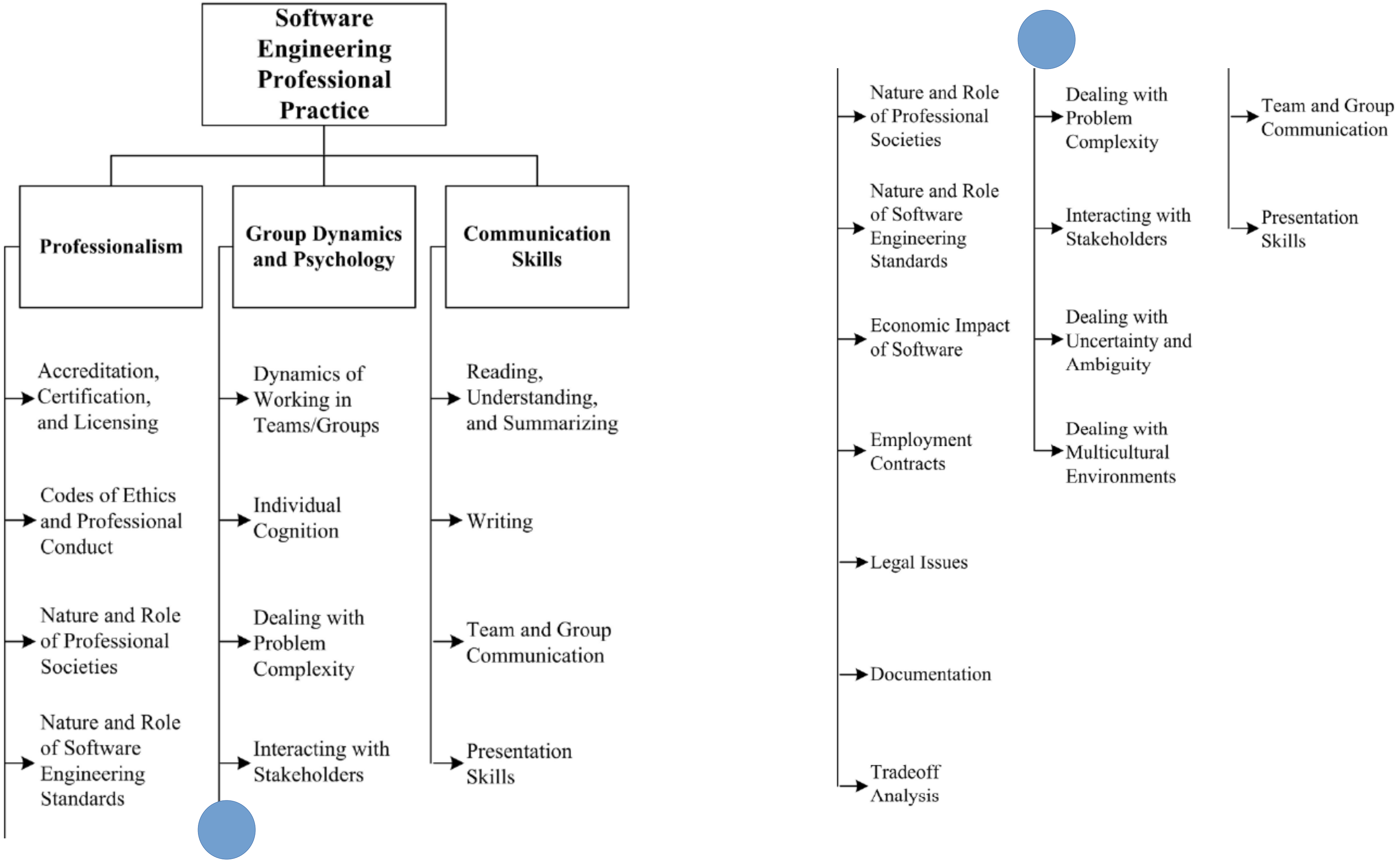
ENGENHARIA DE SOFTWARE:
9. SOFTWARE ENGINEERING MODELS AND METHODS



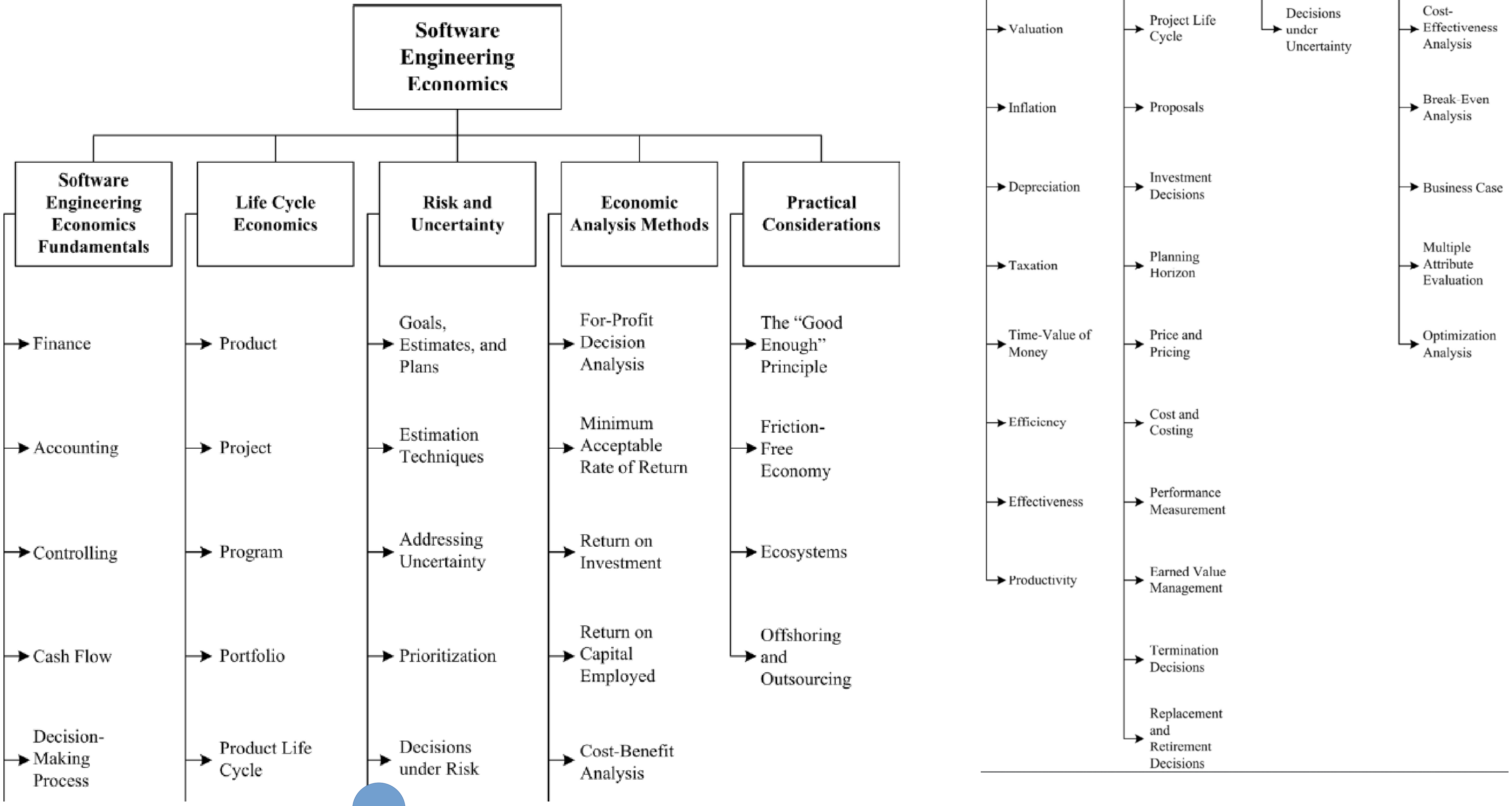
ENGENHARIA DE SOFTWARE:
10. SOFTWARE QUALITY



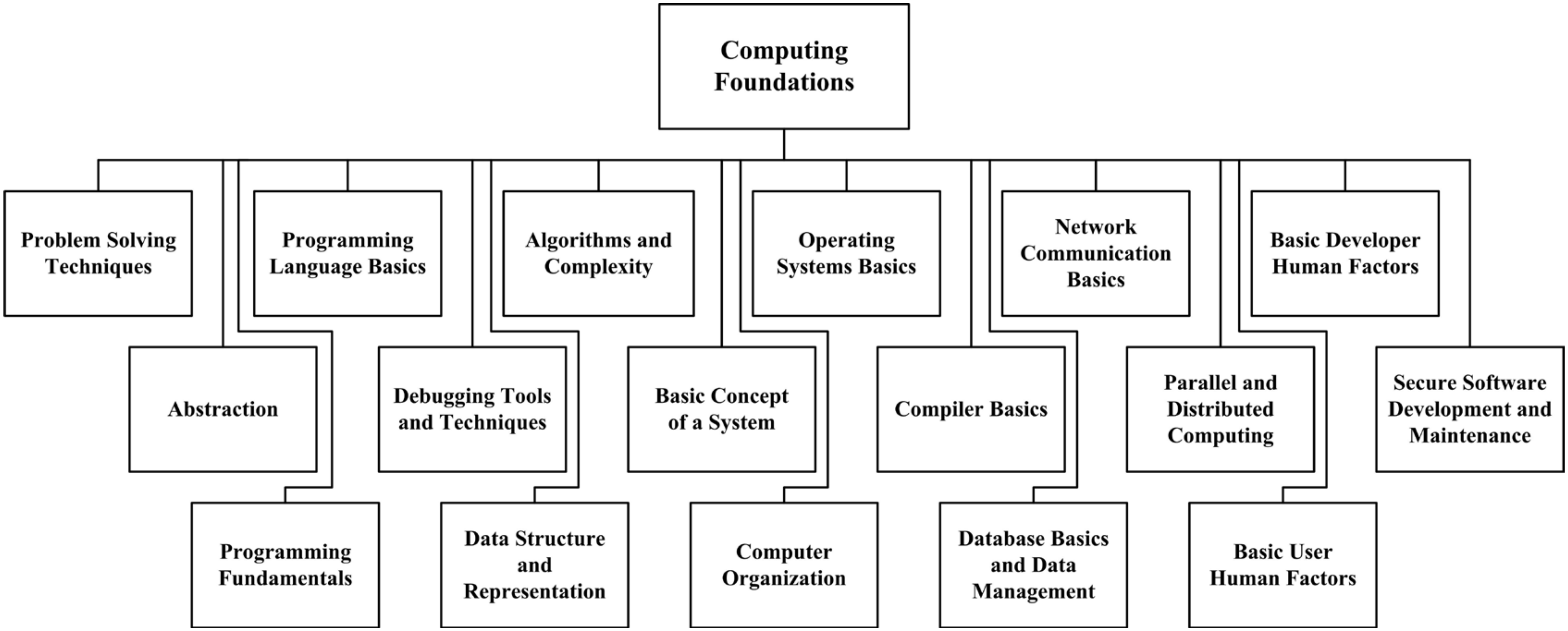
ENGENHARIA DE SOFTWARE:
11. SOFTWARE ENGINEERING PROFESSIONAL PRACTICE



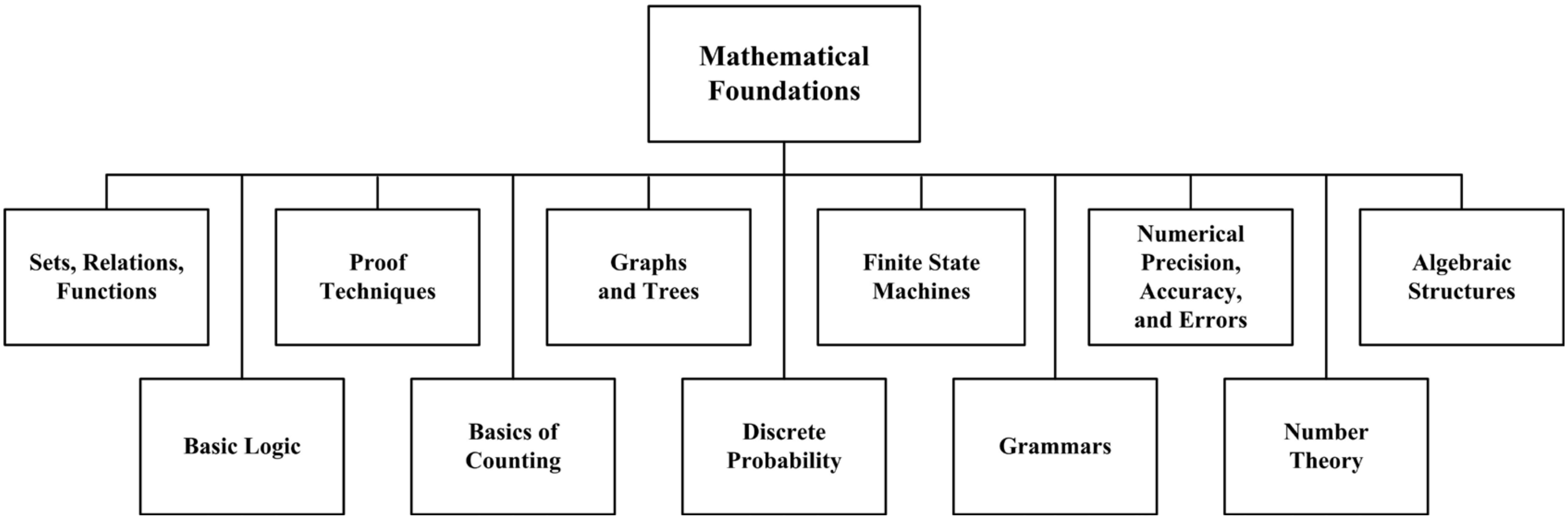
ENGENHARIA DE SOFTWARE:
12. SOFTWARE ENGINEERING ECONOMICS



ENGENHARIA DE SOFTWARE:
13. COMPUTING FOUNDATIONS



ENGENHARIA DE SOFTWARE:
14. MATHEMATICAL FOUNDATIONS



ENGENHARIA DE SOFTWARE:
15. ENGINEERING FOUNDATIONS